WORKED SOLUTION

STEP III 1988, Question 8

Pure • Parabola Tangents and Normals

STEP III 1988, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q8

The key idea

Write every line in terms of the parameter t, find where pairs of lines meet, and put those vertices into the determinant. The determinants look messy, but they are antisymmetric in t_1, t_2, t_3, so each must be a multiple of $(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)$. Simple row and column operations bring that factor out and leave the comparison between the two areas.

(The printed question asks for A_3; this is a misprint for A_2, the area of the normal triangle.)

Solution

Tangent and normal

Differentiating $y^2 = 4ax$ with respect to t along $x = at^2$, $y = 2at$ gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{2at} = \dfrac{1}{t}$. The tangent at $(at^2, 2at)$ is $y - 2at = \frac{1}{t}(x - at^2)$, that is

$$ty = x + at^2.$$

This form also holds when $t = 0$ (the tangent at the vertex is $x = 0$). The normal has gradient $-t$:

$$y + tx = 2at + at^3.$$

We assume $a \neq 0$ throughout.

The tangent triangle

Subtracting the tangent equations at t_i and t_j gives $(t_i - t_j)y = a(t_i^2 - t_j^2)$. Since $t_i \neq t_j$ we may divide, so $y = a(t_i + t_j)$ and then $x = t_i y - at_i^2 = at_i t_j$. So

$$\text{tangents at } t_i, t_j \text{ meet at } (at_i t_j,\ a(t_i + t_j)).$$

The tangents have distinct gradients $1/t_i$ (or one is vertical), so no two are parallel and the triangle exists. Write $s = t_1 + t_2 + t_3$. The vertex opposite P_1 is $(at_2t_3,\ a(s - t_1))$, and similarly for the others. Taking out the factor a from the first two columns, then adding $-s$ times column 3 to column 2 and negating column 2,

$$\det \begin{pmatrix} at_2t_3 & a(s-t_1) & 1 \\ at_1t_3 & a(s-t_2) & 1 \\ at_1t_2 & a(s-t_3) & 1 \end{pmatrix} = -a^2 \det \begin{pmatrix} t_2t_3 & t_1 & 1 \\ t_1t_3 & t_2 & 1 \\ t_1t_2 & t_3 & 1 \end{pmatrix} = a^2 D,$$

where we swapped the first two columns and set

$$D = \det \begin{pmatrix} t_1 & t_2t_3 & 1 \\ t_2 & t_1t_3 & 1 \\ t_3 & t_1t_2 & 1 \end{pmatrix}.$$

To evaluate D, replace row 1 by row 1 − row 2 and row 2 by row 2 − row 3. Row 1 becomes $(t_1 - t_2,\ t_3(t_2 - t_1),\ 0) = (t_1 - t_2)(1, -t_3, 0)$, and row 2 becomes $(t_2 - t_3)(1, -t_1, 0)$. Expanding along the last column,

$$D = (t_1 - t_2)(t_2 - t_3) \det \begin{pmatrix} 1 & -t_3 \\ 1 & -t_1 \end{pmatrix} = (t_1 - t_2)(t_2 - t_3)(t_3 - t_1).$$

Hence

$$A_1 = \tfrac{1}{2} a^2 \,|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|,$$

which is non-zero because the t_i are distinct.

The normal triangle

Subtracting the normal equations at t_i and t_j gives $(t_i - t_j)x = 2a(t_i - t_j) + a(t_i^3 - t_j^3)$, so dividing by $t_i - t_j \neq 0$,

$$x = 2a + a(t_i^2 + t_it_j + t_j^2), \qquad y = 2at_i + at_i^3 - t_ix = -at_it_j(t_i + t_j).$$

The normals have distinct gradients $-t_i$, so again every pair meets. For the vertex opposite P_1, use $t_2 + t_3 = s - t_1$ and write $e_1 = t_2t_3$ (and $e_2 = t_1t_3$, $e_3 = t_1t_2$). Then

$$t_2^2 + t_2t_3 + t_3^2 = (t_2 + t_3)^2 - t_2t_3 = (s - t_1)^2 - e_1, \qquad -t_2t_3(t_2 + t_3) = -se_1 + p,$$

where $p = t_1t_2t_3$. So the vertex is $a(2 + (s - t_1)^2 - e_1,\ -se_1 + p)$, and similarly for the others. Now take out a^2 and do column operations, which don't change the determinant: subtract 2 times column 3 from column 1, subtract p times column 3 from column 2, then take out the factor $-s$ from column 2:

$$\begin{aligned} \frac{1}{a^2} \det(\text{normal vertices}) &= \det \begin{pmatrix} (s-t_i)^2 - e_i & -se_i & 1 \end{pmatrix}_{i=1,2,3} \\ &= -s \det \begin{pmatrix} (s-t_i)^2 - e_i & e_i & 1 \end{pmatrix}_{i=1,2,3}, \end{aligned}$$

where each display shows the typical row i. Next add column 2 to column 1, which removes the $-e_i$, and expand $(s - t_i)^2 = s^2 - 2st_i + t_i^2$. The s^2 is a multiple of column 3 and can be removed, and the determinant is linear in column 1:

$$\det \begin{pmatrix} t_i^2 - 2st_i & e_i & 1 \end{pmatrix} = \det \begin{pmatrix} t_i^2 & e_i & 1 \end{pmatrix} - 2s \det \begin{pmatrix} t_i & e_i & 1 \end{pmatrix} = E - 2sD,$$

say. The second determinant is exactly D.

Ask yourself How do we evaluate E, the determinant with rows $(t_i^2,\ e_i,\ 1)$?

Use the same row operations as for D. Row 1 − row 2 is $(t_1^2 - t_2^2,\ t_3(t_2 - t_1),\ 0) = (t_1 - t_2)(t_1 + t_2,\ -t_3,\ 0)$, and row 2 − row 3 is $(t_2 - t_3)(t_2 + t_3,\ -t_1,\ 0)$. So

$$\begin{aligned} E &= (t_1 - t_2)(t_2 - t_3)[-t_1(t_1 + t_2) + t_3(t_2 + t_3)] \\ &= (t_1 - t_2)(t_2 - t_3)[(t_3^2 - t_1^2) + t_2(t_3 - t_1)] \\ &= (t_1 - t_2)(t_2 - t_3)(t_3 - t_1)(t_1 + t_2 + t_3) = sD. \end{aligned}$$

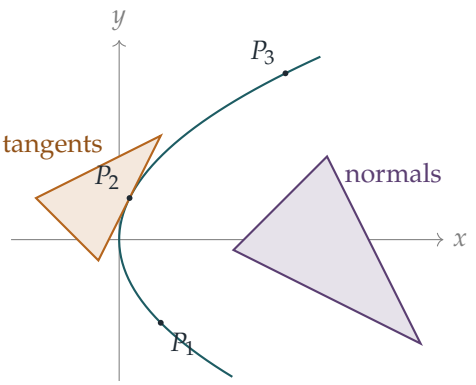
Putting everything together,

$$\det(\text{normal vertices}) = a^2 \cdot (-s)\,(sD - 2sD) = a^2 s^2 D.$$

Taking half the absolute value and comparing with $A_1 = \frac{1}{2}a^2|D|$,

$$A_2 = (t_1 + t_2 + t_3)^2 A_1,$$

which is the required result.



The figure takes $a = 1$ and $t_1, t_2, t_3 = -1, \frac{1}{2}, 2$, so $s = \frac{3}{2}$. The tangent triangle (amber) has area $\frac{27}{8}$, and the normal triangle (plum) has area $\frac{243}{32} = (\frac{3}{2})^2 \cdot \frac{27}{8}$.

When are the normals concurrent?

No two normals are parallel, so they meet in pairs at three points Q_{12}, Q_{13}, Q_{23}. If the normals are concurrent, these points coincide and the triangle has zero area. Conversely, suppose the area is zero but the points are not all equal, say $Q_{12} \neq Q_{13}$. Both points lie on the normal at P_1, so the line through them is that normal. Since the points are collinear, Q_{23} also lies on the normal at P_1. Then Q_{23} lies on all three normals, so $Q_{23} = Q_{12}$ (the only common point of the normals at P_1 and P_2) and likewise $Q_{23} = Q_{13}$, which is a contradiction. So

$$\text{normals concurrent} \iff A_2 = 0 \iff (t_1 + t_2 + t_3)^2 A_1 = 0 \iff t_1 + t_2 + t_3 = 0,$$

using $A_1 \neq 0$ because the t_i are distinct. The required condition is $t_1 + t_2 + t_3 = 0$.

Checks

- With $t_1, t_2, t_3 = 0, 1, 2$ and $a = 1$, the tangent vertices are $(2, 3)$, $(0, 2)$, $(0, 1)$, giving area 1, and $\frac{1}{2}|D| = \frac{1}{2}|(-1)(-1)(2)| = 1$.
- The normals at $t = 1, 2, -3$ (sum zero, $a = 1$) all pass through $(9, -6)$. For example, the normal at $t = 1$ is $y + x = 3$.
- A computer computed the vertices exactly with fractions for about 2,800 random choices of a and distinct t_i, and confirmed $A_2 = (t_1 + t_2 + t_3)^2 A_1$ every time.

Common slips

- Forgetting to justify that the three lines really form a triangle. Pairwise non-parallel is what we need, and here it follows from the distinct gradients.
- Concluding concurrency from $A_2 = 0$ without a reason. Zero area only says the three meeting points are collinear, and the short argument above upgrades this to "equal".
- Expanding the 3×3 determinants in full. This is possible, but sign errors are almost certain. Factor out $(t_i - t_j)$ with row operations first.

Beyond STEP

The factor $(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)$ is a Vandermonde determinant, and the cofactor $t_1 + t_2 + t_3$ in $E = sD$ is a symmetric polynomial. Any antisymmetric polynomial is the Vandermonde product times a symmetric one, a fact that reappears in the theory of determinants and symmetric functions. The triangle of normals is related to the *evolute* of the parabola, the curve traced out by its centres of curvature, which is the envelope of the normals. Strang's *Introduction to Linear Algebra* covers determinants and their row operations well.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.